

Obligatory Exercise 1

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Exercise O1.1

a) $F_1 : ((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$

$$\begin{array}{c}
 \frac{q \rightarrow r, p \Rightarrow r, p}{p \rightarrow q, q \rightarrow r, p \Rightarrow r} \text{ax} \quad \frac{q, p \Rightarrow r, q}{q, q \rightarrow r, p \Rightarrow r} \text{ax} \quad \frac{q, r, p \Rightarrow r}{\rightarrow -l} \text{ax} \\
 \frac{\frac{\frac{p \rightarrow q, q \rightarrow r, p \Rightarrow r}{(p \rightarrow q) \wedge (q \rightarrow r), p \Rightarrow r} \wedge -l}{(p \rightarrow q) \wedge (q \rightarrow r) \Rightarrow p \rightarrow r} \rightarrow -r}{\Rightarrow ((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)} \rightarrow -r
 \end{array}$$

As you can see all the branches were closed successfully where we got $\dots, p \Rightarrow \dots, p$ on the left branch, $q, \dots \Rightarrow \dots, q$ on the middle branch, and $\dots, r, \dots \Rightarrow r$ on the right branch. Next on the list is F_2 .

$F_2 : \forall x \exists y (p(x) \wedge (p(y) \rightarrow q(x))) \rightarrow \forall z q(z)$

$$\begin{array}{c}
 \frac{\dots, p(b), p(c) \rightarrow q(b), p(a) \Rightarrow q(a), p(b)}{\dots, p(b) \wedge (p(c) \rightarrow q(b)), p(a) \Rightarrow q(a), p(b)} \text{ax} \\
 \frac{\dots, p(b) \wedge (p(c) \rightarrow q(b)), p(a) \Rightarrow q(a), p(b)}{\dots, \exists y (p(b) \wedge (p(y) \rightarrow q(b))), p(a) \Rightarrow q(a), p(b)} \wedge -l \\
 \frac{\dots, \exists y (p(b) \wedge (p(y) \rightarrow q(b))), p(a) \Rightarrow q(a), p(b)}{\dots, p(a) \Rightarrow q(a), p(b)} \exists -l \\
 \frac{\dots, p(a) \Rightarrow q(a), p(b)}{\dots, p(a) \Rightarrow q(a), p(b)} \forall -l \quad \frac{\dots, p(a), q(a) \Rightarrow q(a)}{\rightarrow -l} \text{ax} \\
 \frac{\dots, p(a), p(b) \rightarrow q(a) \Rightarrow q(a)}{\dots, p(a) \wedge (p(b) \rightarrow q(a)) \Rightarrow q(a)} \wedge -l \\
 \frac{\dots, p(a) \wedge (p(b) \rightarrow q(a)) \Rightarrow q(a)}{\dots, \exists y (p(a) \wedge (p(y) \rightarrow q(a))) \Rightarrow q(a)} \exists -l \\
 \frac{\dots, \exists y (p(a) \wedge (p(y) \rightarrow q(a))) \Rightarrow q(a)}{\forall x \exists y (p(x) \wedge (p(y) \rightarrow q(x))) \Rightarrow q(a)} \forall -l \\
 \frac{\forall x \exists y (p(x) \wedge (p(y) \rightarrow q(x))) \Rightarrow q(a)}{\Rightarrow \forall x \exists y (p(x) \wedge (p(y) \rightarrow q(x))) \rightarrow \forall z q(z)} \forall -r \\
 \frac{\Rightarrow \forall x \exists y (p(x) \wedge (p(y) \rightarrow q(x))) \rightarrow \forall z q(z)}{\Rightarrow \forall x \exists y (p(x) \wedge (p(y) \rightarrow q(x))) \rightarrow \forall z q(z)} \rightarrow -r
 \end{array}$$

As you can see, once again all the branches were closed successfully where $\dots, p(b), \dots \Rightarrow \dots, p(b)$ on the left branch and $\dots, q(a) \Rightarrow q(a)$ on the right branch.

b) First we attempt to prove F_3 using the Sequent Calculus LK.
 $F_3 : ((p \rightarrow r) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow q)$

$$\frac{\frac{\frac{q \rightarrow r, p \Rightarrow q, p}{p \rightarrow r, q \rightarrow r, p \Rightarrow q} \text{ ax} \quad \frac{\frac{r, p \Rightarrow q, q}{r, q \rightarrow r, p \Rightarrow q} \quad \frac{r, r, p \Rightarrow q}{r, q \rightarrow r, p \Rightarrow q} \rightarrow -l}{\frac{p \rightarrow r, q \rightarrow r, p \Rightarrow q}{(p \rightarrow r) \wedge (q \rightarrow r), p \Rightarrow q} \wedge -l} \rightarrow -l$$

$$\frac{\frac{(p \rightarrow r) \wedge (q \rightarrow r), p \Rightarrow q}{(p \rightarrow r) \wedge (q \rightarrow r) \Rightarrow p \rightarrow q} \rightarrow -r}{\Rightarrow ((p \rightarrow r) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow q)} \rightarrow -r$$

After failing we see exactly where the problem arises. We can not close the $r, p \Rightarrow q$ branch. Now, we can easily deduce and create a counter model. We start by assigning the values p, q , and r according to our unsaleable branch: $V_I(p) = \top, V_I(q) = \perp, V_I(r) = \top$, after which we evaluate the formula to verify.

- 1) $V_I(p \rightarrow r) = \top \rightarrow \top = \top$
 $V_I(q \rightarrow r) = \perp \rightarrow \top = \top$
 $V_I(p \rightarrow q) = \top \rightarrow \perp = \perp$
- 2) $V_I((p \rightarrow r) \wedge (q \rightarrow r)) = \top \wedge \top = \top$
- 3) $V_I(((p \rightarrow r) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow q)) = \top \rightarrow \perp = \perp$

For a formula to be valid it always has to evaluate to $\top$ (true). However, we just witnessed an example of the formula evaluating to $\perp$ (false) which makes it falsifiable or in other words, invalid.

Exercise O1.2

a) These are the rules for $\uparrow -l$ and $\uparrow -r$:

$$\frac{\Gamma \Rightarrow A, \Delta \quad \Gamma \Rightarrow B, \Delta}{\Gamma, A \uparrow B \Rightarrow \Delta} \uparrow -l$$

$$\frac{\Gamma, A, B \Rightarrow \Delta}{\Gamma \Rightarrow A \uparrow B, \Delta} \uparrow -r$$

These rules can be deduced from the negation and conjunction rules. That is because as defined before $A \uparrow B \equiv \neg(A \wedge B)$. Thus, by applying a negation rule first, and a conjunction rule afterwards we can arrive to the solution above.

Here is the process for $\uparrow -l$:

$$\frac{\frac{\Gamma \Rightarrow A, \Delta \quad \Gamma \Rightarrow B, \Delta}{\Gamma \Rightarrow \Delta, A \wedge B} \wedge - r}{\Gamma, \neg(A \wedge B) \Rightarrow \Delta} \neg - l$$

And here is one for $\uparrow -r$:

$$\frac{\frac{\Gamma, A, B \Rightarrow \Delta}{\Gamma, A \wedge B \Rightarrow \Delta} \wedge - l}{\Gamma \Rightarrow \Delta, \neg(A \wedge B)} \neg - r$$

Note that we substituted $A \uparrow B$ with $\neg(A \wedge B)$ to begin with.

b) The central part is the first lemma where the LK-rule must preserve falsifiability upwards. That and other 2 lemmas but these are out of scope in this task.

Now onto proving the first lemma of $\uparrow -l$. First we assume that our interpretation I falsifies the conclusion, all formulae in Δ , and satisfies $\Gamma \cup \{A \uparrow B\}$. Since I satisfies $A \uparrow B$ per model semantics either case 1: $I \not\models A$ or case 2: $I \not\models B$. Therefore, $I \models \Gamma$ and I falsifies either left ($\Delta \cup A$), or right ($\Delta \cup B$).

Now proving the $\uparrow -r$. Again, we assume that I falsifies the conclusion. Then, that $I \models \Gamma$, and $I \not\models \Delta \cup \{A \uparrow B\}$. Since I falsifies $A \uparrow B$, per model semantics $I \models A \cup B$. Thus, $I \models \Gamma \cup A \cup B$ and falsifies all formulae in Δ .